



Measures of Central Tendency

- **Central Tendency:**

- The statistical measure that identifies a single value as representative of an entire distribution. It aims to provide an accurate description of the entire data. It is the single value that is most typical/representative of the collected data.

- **Requisite of Good Measure of Central Tendency**

- Simple to understand
- Simple to compute
- Should be based on all the observations of the data
- Capable of further mathematical treatments
- Not be affected by extreme values

Arithmetic Mean :

The most common measure of central tendency is the arithmetic mean. In layman's terms, the mean of data indicates an average of the given collection of data. It is equal to the sum of all the values in the group of data divided by the total number of values.

For n values in a set of data namely as $x_1, x_2, x_3, \dots, x_n$, the mean of data is given as:

$$\bar{x} = \frac{x_1 + x_2 + x_3 \dots \dots \dots x_n}{n}$$

It can also be denoted as:

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

MEDIAN

Median is the middle value of the given list of data when arranged in an order. The arrangement of data or observations can be made either in ascending order or descending order.

Odd Number of Observations

If the total number of observations given is odd, then the formula to calculate the median is:

$$\text{Median} = \left(\frac{n+1}{2}\right)^{\text{th}} \text{ term}$$

where n is the number of observations

Even Number of Observations

If the total number of observation is even, then the median formula is:

$$\text{Median} = \frac{\left(\frac{n}{2}\right)^{\text{th}} \text{ term} + \left(\frac{n}{2} + 1\right)^{\text{th}} \text{ term}}{2}$$

where n is the number of observations

MODE

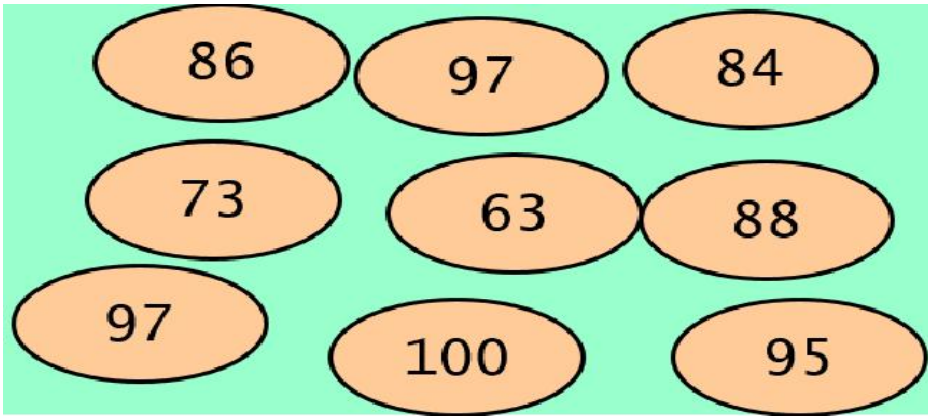
Mode is defined as the value of the variable which occurs more frequently in the data. It is the value which is repeated maximum number of times or with highest frequency

Types of mode

- Unimodal Mode - A set of data with one Mode is known as a Unimodal Mode.
- Bimodal Mode - A set of data with two Modes is known as a Bimodal Mode. This means that there are two data values that are having the highest frequencies.

- Mode- The value in the set of data that occurs most frequently.
- The set of data can have-
 1. One mode
 2. More than one mode
 3. No mode

Example 1 : These are the marks obtained by the class out of hundred. Find Mean, median and mode for the given data set.



Mean : 87

Median : 88

Mode: 97

Find Mean Median and Mode

9, 10, 10, 13, 13

MEAN: 11

Median: 10

Mode: 10, 13

8, 8, 9, 10, 10, 12, 12, 13, 17

Find Mean Median and Mode

$$\begin{array}{r} 11 \\ \hline 10 \\ \hline 8, 10, 12 \end{array}$$

Compute median for the following data 5,20,18,12,21,18,26,5,15,20

The ordered observation are :

5,5,12,15,18,18,20,20,21,26

N=10 i.e. n is even

Median = observation at $(n/2)$ th position + observation at $(n/2 + 1)$ th position

2

=observation at 5th position + observation at 6th position

2

= 18+18

2

=18

Example 6:

Two variables X, Y assumes the values $X_1 = 2$, $X_2 = -5$, $X_3 = 4$, $X_4 = -8$ and $Y_1 = -3$, $Y_2 = -8$, $Y_3 = 10$, $Y_4 = 6$.

Calculate- $\sum XY$, $\sum X \sum Y$, $\sum XY^2$, $\sum X^2$, $\sum (X - Y)(X + Y)$

ARITHMETIC MEAN:

- Arithmetic Mean for raw data
- If there are n observations in the set and they represent different values $x_1, x_2, \dots, x_n$ of the variable X Then the A.M. is denoted by
- $\bar{X} = \text{Sum of observations} / \text{Number of observations}$

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n} = \frac{\sum x}{n}$$

Calculate A.M for the following data representing marks scored by 7 students

Marks are as follows

610, 680, 690, 630, 700, 600, 780

$$\bar{X} = 670$$

Arithmetic Mean For Ungrouped Frequency Distribution

- The variable X assumes distinct values $x_1, x_2, \dots, x_n$ with corresponding frequencies $f_1, f_2, \dots, f_n$.
- The Arithmetic Mean is given by

$$\bar{x} = \frac{f_1x_1 + f_2x_2 + f_3x_3 + \dots + f_nx_n}{N} = \frac{\sum fx}{N}$$

where

$$N = \sum f$$

Example :

The following data give weight distribution of students in a class

Weight	41	42	43	44	45	46	47	48
No of Students	3	6	8	15	17	12	5	4

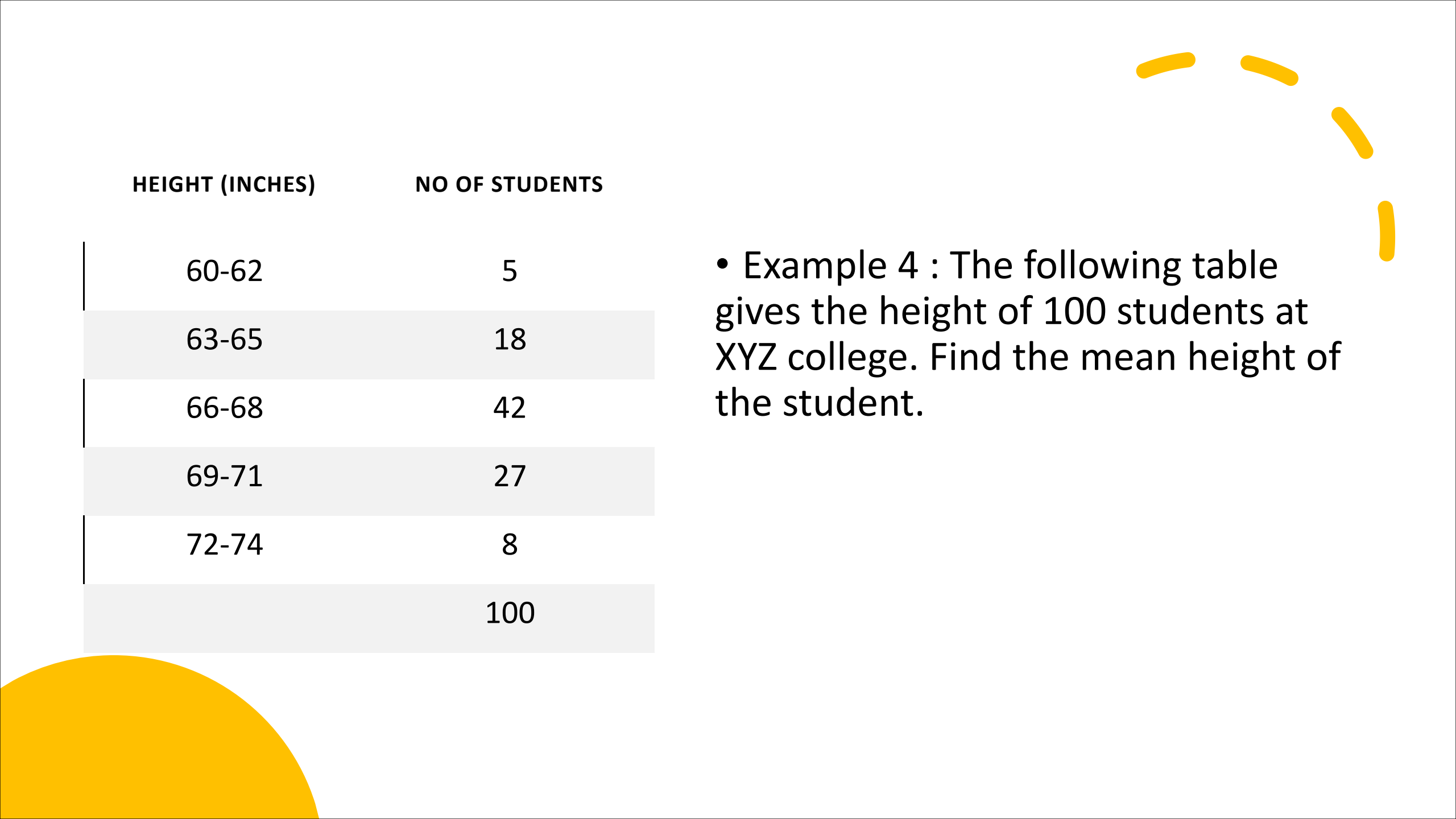
Mean: 44.61

Arithmetic mean For grouped Frequency Distribution

Find A.M for the data below

Marks	0-10	10-20	20-30	30-40	40-50	50-60
No of Students	5	10	2	30	20	10

- 
- Example 2: If the mean of 6, 4, 7, p and 10 is 8, find the value of p .
 - Example 3: The mean of 40 observations was 160. It was detected on rechecking that the value of 165 was wrongly copied as 125. Find the correct mean.



HEIGHT (INCHES)	NO OF STUDENTS
60-62	5
63-65	18
66-68	42
69-71	27
72-74	8
	100

- Example 4 : The following table gives the height of 100 students at XYZ college. Find the mean height of the student.

Class Interval	Height (inches)	No of students
	60-62	5
	63-65	18
	66-68	42
	69-71	27
	72-74	8
		100

L1 = Lower class Limit

L2 = Upper Class Limit

Solution:

AM for grouped data:

$$\bar{x} = \frac{\sum(f * x)}{\sum f}$$

Class Mark:

Midpoint of class is give as

$$x = \frac{l_1 + l_2}{2}$$



Weekly Wage (\$)	No of employees
250-259	8
260-269	10
270-279	16
280-289	14
290-299	10
300-309	5
310-319	2

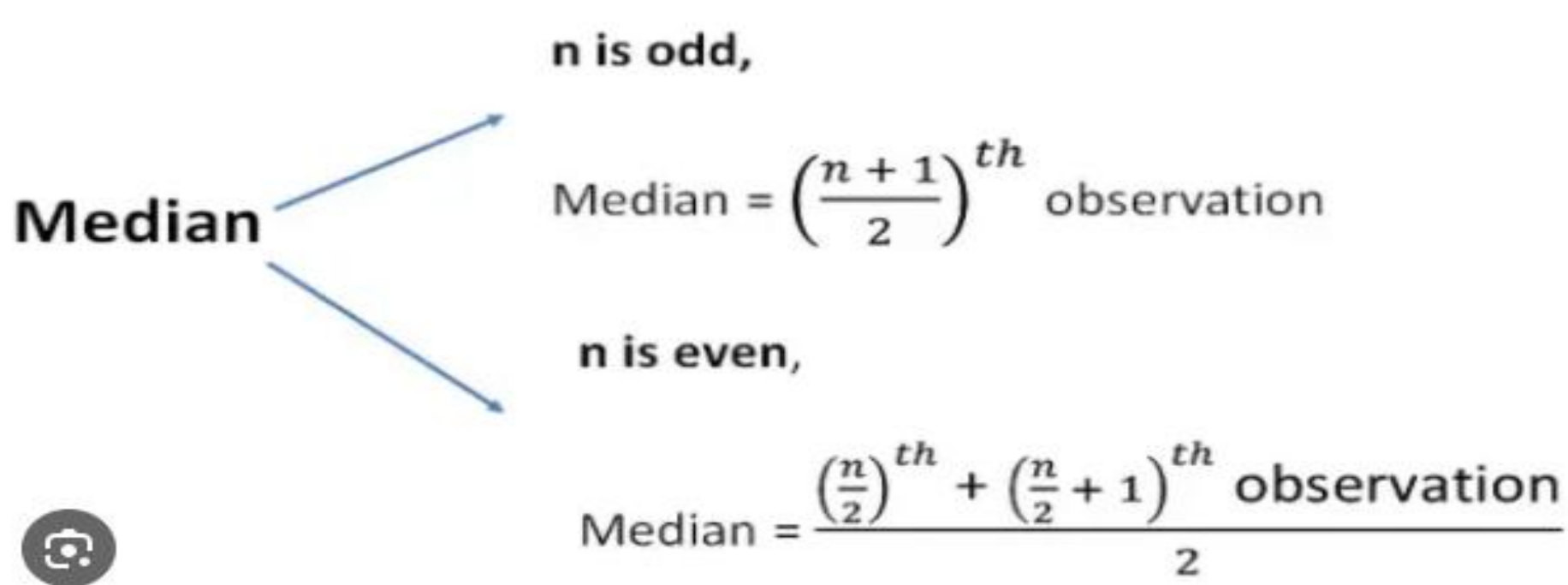
- Example 5 : Use the following frequency distribution of weekly wages to find A.M. of wage of employees at P & R company.

Median:

Median is the value of central observation in the data when the observations are arranged in increasing order of magnitude. It divides data into two parts containing equal number of observations above and below it.

i. Median for Raw data

Suppose there are n observations in the data. Arrange these observations in ascending order or magnitude. Median is then the central observation of the data



Following are the temperatures recorded in certain city during a week :
35,38,40,39,37,35,36. Find the median temperature

The ordered observation are
35,35,36,37,38,39,40

Here $n=7$ n is odd

Median = observation at $\left(\frac{n+1}{2}\right)^{th}$ position in the ordered data

= observation at 4th position

=37

COMPUTE MEDIAN FOR THE FOLLOWING SERIES

5,20,18,12,21,18,26,5,15,20

SOLUTION:

The ordered observations are 5,5,12,15,18,18,20,20,21,26

Here n is even i.e. n=10

Median = $\frac{\text{observation at } \left(\frac{n}{2}\right)^{th} \text{ position} + \text{observation at } \left(\frac{n}{2} + 1\right)^{th} \text{ position}}{2}$

= $\frac{\text{observation at 5th position} + \text{observation at 6th position}}{2}$

= $(18+18)/2$

= 18

MEDIAN FOR UNGROUPED FREQUENCY DISTRIBUTION

1. Find less than type cumulative frequency
2. Find total frequencies $N = \sum f$
3. If N is odd find the value of $\left(\frac{N+1}{2}\right)^{th}$. The median is that value of the variable at which less than type cumulative frequency is equal to or exceeds $\left(\frac{N+1}{2}\right)^{th}$ for the first time
4. If N is even find 2 values of the variable at which less than type cumulative frequency is equal to or exceeds $\left(\frac{N}{2}\right)$ and $\left(\frac{N}{2} + 1\right)$. If these 2 values are same this value is the median of the distribution. If 2 values are different the median is the arithmetic mean of 2 values

Find median for the following data

Weight	12	13	14	15	16	17
No of students	5	14	20	16	8	4

Solution :

Age (in years) (x)	No. of Students (f)	Cumulative Frequency < Type
12	5	5
13	14	19
14	20	39
15	16	55
16	8	63
17	4	67
	$N = 67$	

$$N = 67 \text{ i.e. odd.} \quad \therefore \frac{N + 1}{2} = \frac{67 + 1}{2} = 34$$

Less than type cumulative frequency exceeds the value 34 for the first time when $x = 14$

$\therefore$ Median = 14 years.

x	2	3	4	5	6	7
f	2	8	5	15	7	3

Find median for the following frequency distribution :

<i>x</i>	2	3	4	5	6	7
<i>f</i>	2	8	15	15	7	3

Solution :

<i>x</i>	<i>f</i>	Less than type Cumulative frequency
2	2	2
3	8	10
4	15	25
5	15	40
6	7	47
7	3	50
	<i>N</i> = 50	

Check the Less than type cf value greater than 25 and less than type cf value greater than 26

- Less than type cf ≥ 25 is for $x=4$
- Less than type cf ≥ 26 is for $x=5$

$N = 50$ i.e. even. $\therefore \frac{N}{2} = 25$ and $\frac{N}{2} + 1 = 26$

From less than type cumulative frequencies, the 25th observation is 4 and 26th observation will be 5.

$$\therefore \text{Median} = \frac{4 + 5}{2} = 4.5$$

Median for the grouped data

Median Class- The class interval for which cumulative frequency is just greater than $N/2$

$$\text{Median} = l_1 + \left\{ \frac{(l_2 - l_1)}{f} \left(\frac{N}{2} - c.f. \right) \right\}$$

Where,

l_1 = lower class boundary of the median class

l_2 = upper class boundary of the median class

f = frequency of median class

$c.f.$ = cumulative frequency preceding the median class

$N = \sum f$

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$$N = \sum f$$

Weight (pounds)	No of students
118-126	3
127-135	5
136-144	9
145-153	12
154-162	5
163-171	4
172-180	2

Class	Frequency f	c.f.
118-126	3	3
127-135	5	8
136-144	9	17
145-153	12	29
154-162	5	34
163-171	4	38
172-180	2	40
	$\Sigma f = 40$	

$$N = \Sigma f = 40$$

$$\frac{N}{2} = 20$$

Locate class for which c.f. is just greater than $\frac{N}{2}$

Median Class

$$L1 = 145$$

$$L2 = 153$$

$$F = 12$$

$$c.f. = 17$$

Calculate median age from the following frequency distribution :

Age (in years)	0-9	10-19	20-29	30-39	40-49
Frequency	3	10	28	36	3

Solution :

Here, the given class-intervals are inclusive type. They are first converted into continuous type as follows :

Age (in years)	Number of Persons	Less than type Cumulative Frequency
0-9.5	3	3
9.5-19.5	10	13
19.5-29.5	28	41
29.5-39.5	36	79
39.5-49.5	3	80
	$N = 80$	

$$\frac{N}{2} = \frac{80}{2} = 40$$

The 'less than type' cumulative frequency which exceeds $\frac{N}{2} = 40$ for the first time is 41.

$\therefore$ The median class is (19.5 - 29.5)

$\therefore l_1 = 19.5, l_2 = 29.5, c.f. = 13, f = 28$

$$\begin{aligned}\therefore M &= l_1 + \frac{(l_2 - l_1) \left(\frac{N}{2} - c.f. \right)}{f} \\ &= 19.5 + \frac{(29.5 - 19.5) (40 - 13)}{28} \\ &= 19.5 + \frac{10 \times 27}{28} \\ &= 29.1\end{aligned}$$

$\therefore$ Median age is 29.1 years.

Mode for the grouped data

- Modal Class- The class interval with maximum frequency

$$\text{Mode} = l_1 + \left\{ (l_2 - l_1) \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \right\}$$

Where,

l_1 = lower class boundary of the modal class

l_2 = upper class boundary of the modal class

f_0 = frequency of the class preceding modal class

f_1 = frequency of the modal class

f_2 = frequency of the class succeeding modal class

Find the mode of the data given below

Viewing time (mins)	No of students
9.3-9.7	2
9.8-10.2	5
10.3-10.7	12
10.8-11.2	17
11.3-11.7	14
11.8-12.2	6
12.3-12.7	3
12.8-13.2	1

Viewing time (mins)	No of students f
9.3-9.7	2
9.8-10.2	5
10.3-10.7	12 f_0
10.8-11.2	17 f_1
11.3-11.7	14 f_2
11.8-12.2	6
12.3-12.7	3
12.8-13.2	1

$$\text{Mode} = l_1 + \left\{ (l_2 - l_1) \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \right\}$$

Modal Class- The class interval with maximum frequency

Frequency preceding modal class

Highest frequency

Frequency succeeding modal class

Modal Class

$l_1 = 10.8$

$l_2 = 11.2$

Mode =

Example 29 :

Calculate modal income for the following income distribution :

Daily Income (₹)	below 30	31-60	61-90	91-120	121-150	Above 150
No. of Persons	22	110	198	95	42	33

We first convert the given distribution into continuous form as follows :

Class-boundaries	Frequency
below 30.5	22
30.5-60.5	110
60.5-90.5	198
90.5-120.5	95
120.5-150.5	42
Above 150.5	33

The highest frequency is 198.

∴ The modal class is (60.5 – 90.5)

∴ $l_1 = 60.5$, $l_2 = 90.5$, $f_1 = 198$, $f_0 = 110$, $f_2 = 95$

$$\begin{aligned}\text{Mode} &= l_1 + \frac{(l_2 - l_1)(f_1 - f_0)}{(f_1 - f_0) + (f_1 - f_2)}, \\ &= 60.5 + \frac{30 \times (198 - 110)}{(198 - 110) + (198 - 95)} \\ &= 74.32\end{aligned}$$

∴ Modal daily income is ₹ 74.32.

PARTITION VALUES : QUARTILES,DECILES,PERCENTILES

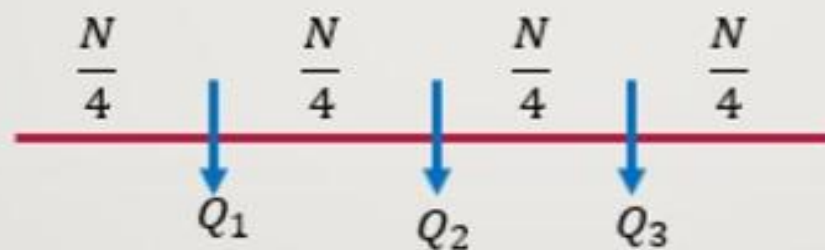
Suppose, N observations are arranged in order of magnitude.

- **Median**- The value which divides data in 2 equal parts.
- **Quartiles**- The values which divide data set into 4 equal parts. Denoted by Q_1, Q_2, Q_3 and are called as 1st, 2nd and 3rd quartiles respectively.
- **Deciles**- The values which divide data into 10 equal parts. Denoted by : $D_1 \dots D_9$ and are called as 1st, 2nd, 9th deciles respectively.
- **Percentiles**- The values which divide data into 100 equal parts. Denoted by $P_1 \dots P_{99}$ and are called as 1st, 2nd, 99th percentiles respectively.

Median-



Quartiles-



Deciles-



Percentiles-



Quartiles for Raw Data:

In case of individual observations , quartiles can be determined as follows:
Suppose there are n observations in the data. Arrange these observations in ascending order of magnitude. Then

. First quartile (Q_1) = $\left(\frac{n+1}{4}\right)^{\text{th}}$ term

. Second quartile (Q_2) = $\left(\frac{n+1}{2}\right)^{\text{th}}$ term

. Third quartile (Q_3) = $\left(\frac{3(n+1)}{4}\right)^{\text{th}}$ term

E.g. Find Q1 and Q3 for the following data
50,35,48,68,87,61,44,56,74

SOLUTION:

The given data in ascending order is
35,44,48,50,56,67,68,74,87

$$\begin{aligned} Q1 &= (n+1)/4 \text{ observation} = (2.5)^{\text{th}} \text{ observation} = (2^{\text{nd}} \text{ observation} + 3^{\text{rd}} \text{ observation})/2 \\ &= (44+48)/2 \\ &= 46 \end{aligned}$$

$$\begin{aligned} Q3 &= 3(n+1)/4 \text{ observation} = 7.5^{\text{th}} \text{ observation} = (7^{\text{th}} \text{ obs} + 8^{\text{th}} \text{ observation})/2 \\ &= 68+74/2 \\ &= 71 \end{aligned}$$

Quartiles for ungrouped frequency distribution

1. Find $i(N+1)/4$; N = Total Frequency
2. Find less than type cumulative frequency for the given ungrouped frequency distribution
3. Q_i is that value of the variable where less than type cumulative frequency is equal or just greater than $i(N+1)/4$

X	10	20	30	40	50	60	70	80
f	15	30	53	75	100	110	115	125

Quartiles for Grouped Frequency Distribution

1. Find less than type cumulative frequency
2. Locate i th quartile class as the class interval where less than cf is equal to or greater than the value of $iN/4$ for the first time, N is the total frequency
3. The value of i^{th} quartile can then be calculated using formula

$$Q_i = l_1 + \frac{(l_2 - l_1) \left(\frac{iN}{4} - c.f. \right)}{f}, \quad i = 1, 2, 3;$$

where,

l_1 : lower class-boundary of the i^{th} quartile class

l_2 : upper class-boundary of the i^{th} quartile class.

$c.f.$: less than type cumulative frequency of the class-interval just preceding the i^{th} quartile class.

f : frequency of the i^{th} quartile class.

Note : Q_1 and Q_3 are the Limits within which middle 50% of the observations lie.

DECILES

1. Find less than type cumulative frequency
2. Locate i th quartile class as the class interval where less than cf is equal to or greater than the value of $iN/10$ for the first time, N is the total frequency
3. The value of i^{th} quartile can then be calculated using formula

$$D_i = l_1 + \frac{(l_2 - l_1) \left(\frac{iN}{10} - c.f. \right)}{f}, i = 1, 2, \dots, 9;$$

where,

l_1 : lower class-boundary of the i^{th} decile class

l_2 : upper class-boundary of the i^{th} decile class.

$c.f.$: less than type cumulative frequency of the class-interval just preceding the i^{th} decile class.

f : frequency of the i^{th} decile class.

The ninety nine percentiles are denoted as $P_1, P_2, \dots, P_{99}$. The i^{th} percentile P_i can be calculated for a grouped frequency distribution as

PERCENTILES

1. Find less than type cumulative frequency
2. Locate i th quartile class as the class interval where less than type cf is equal to or greater than the value of $iN/100$ for the first time, N is the total frequency
3. The value of i^{th} quartile can then be calculated using formula

$$P_i = l_1 + \frac{(l_2 - l_1) \left(\frac{iN}{100} - c.f. \right)}{f},$$

where,

l_1 : lower class-boundary of the i^{th} percentile class

l_2 : upper class-boundary of the i^{th} percentile class.

$c.f.$: less than type cumulative frequency of the class-interval just preceding the i^{th} percentile class.

f : frequency of the i^{th} percentile class.

For the following data, calculate 1st Quartile, 3rd Decile and 54th percentile.

Age	No of people
0-10	6
10-20	8
20-30	13
30-40	18
40-50	16
50-60	13
60-70	12
70-80	9
80-90	4
90-100	1

$$Q1=28.4615$$

$$Q2=43.125$$

$$Q3=60.833$$

$$D3=31.1667$$

$$P54=45.625$$

Age	No of People	less than type cf
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0-10	6	6
10-20	8	14
20-30	13	27
30-40	18	45
40-50	16	61
50-60	13	74
60-70	12	86
70-80	9	95
80-90	4	99
90-100	1	100

N=100

1st Quartiles

less for Q_1 (1st Quartile)

$$\frac{N}{4} = \frac{100}{4} = 25$$

less than type cf ≥ 25 is 27
The quartile class is 20-30

$$l_1 = 20 \quad l_2 = 30 \quad f = 13 \quad cf = 14$$

$$Q_1 = l_1 + \frac{(l_2 - l_1) \left(\frac{N}{4} - cf \right)}{f}$$

$$= 20 + \frac{10 (25 - 14)}{13}$$

$$= 28.4615$$

3rd Decile

For 3rd Decile

$$D_i = \frac{iN}{10}$$

$$D_3 = \frac{3 \times 100}{10} = 30$$

Less than type $cf \geq 30$ is 45

The decile class is 30-40

$l_1 = 30$ $l_2 = 40$ $f = 18$ $cf = 27$

$$D_3 = l_1 + \frac{(l_2 - l_1) \left(\frac{3N}{10} - cf \right)}{f}$$

$$= 30 + \frac{(40 - 30)(30 - 27)}{18}$$

$$= 30 + \frac{(10)(3)}{18}$$

$$= 31.667$$

54th Percentile

For 54th Percentile

$$P_i = \frac{iN}{100}$$

$$P_{54} = \frac{54 \times 100}{100} = 54$$

Less than type $cf \geq 54$ is 61

The percentile class is 40-50

$l_1 = 40$ $l_2 = 50$ $f = 18$ $cf = 45$

$$P_{54} = l_1 + \frac{(l_2 - l_1) \left(\frac{54N}{100} - cf \right)}{f}$$

$$= 40 + \frac{(50 - 40)(54 - 45)}{18}$$

$$= 40 + \frac{10 \times 9}{18}$$

$$= 45.625$$

For the following data, calculate Q1 and Q3.

Marks less than	10	20	30	40	50	60	70
No of students	12	30	54	76	96	101	102

Find Q1,Q2 , P50,D8

Daily Expenses	0-100	100-200	200-300	300-400	400-500
Number of Days	3	10	28	30	9

Daily Expenses	Number of Days	Less than type cf
0-100	3	3
100-200	10	13
200-300	28	41
300-400	30	71
400-500	9	80
<u>N = 80</u>		

For Q_1
 $\frac{1N}{4}$

For Q_1 $\frac{1N}{4} = \frac{80}{4} = 20$

Less than type cf ≥ 20 is 41
 The quartile class is 200-300

$$Q_1 = l_1 + \frac{(l_2 - l_1) \left(\frac{N}{4} - cf \right)}{f}$$

$l_1 = 200$ $l_2 = 300$ $f = 28$ $cf = 13$

$$Q_1 = 200 + \frac{(300 - 200)(20 - 13)}{28}$$

$Q_1 = 225$

For Q_2

$$\frac{2N}{4} = \frac{2 \times 80}{4} = 40$$

Less than type cf ≥ 40 is 41
 The quartile class is 200-300

$$Q_2 = l_1 + \frac{(l_2 - l_1) \left(\frac{2N}{4} - cf \right)}{f}$$

$l_1 = 200$ $l_2 = 300$ $f = 28$ $cf = 13$

$$Q_2 = 200 + \frac{(300 - 200)(40 - 13)}{28}$$

$$Q_2 = 296.428$$

For D_1 : $\frac{1N}{10}$

For D_8 :- $\frac{8N}{10}$

$$= \frac{8 \times 80}{10}$$

$$= 64$$

Less than type cf ≥ 64 is 71

The decile class is 300-400

$$l_1 = 300 \quad l_2 = 400 \quad f = 30 \quad cf = 41$$

$$D_8 = l_1 + \frac{(l_2 - l_1) \left(\frac{8N}{10} - cf \right)}{f}$$
$$= 300 + \frac{(400 - 300) (64 - 41)}{30}$$

$$D_8 = 376.67$$

$$\text{For } P_1 :- \frac{1N}{100} \quad \text{For } P_{50} :- \frac{50N}{100}$$

$$P_{50} = \frac{50 \times 80}{100}$$
$$= 40$$

Less than type cf ≥ 40 is 2041

The percentile class is 200-300

$$l_1 = 200 \quad l_2 = 300 \quad f = 28 \quad cf = 13$$

$$P_{50} = l_1 + \frac{(l_2 - l_1) \left(\frac{50N}{100} - cf \right)}{f}$$

$$= 200 + \frac{(300 - 200) (40 - 13)}{28}$$

$$P_{50} = 296.4286$$

Geometric Mean

- It is used for growth rates like population growth or interest rates and percentages.
- Useful in cases when the numbers in the series are not independent of each other.
- Calculating average growth rates, Suppose a stock that grows by 10% in one year, declines by 20% in year 2 and then grows by 30% in year 3
- If bacteria increases its population by 20% in the first year, 30% in the next hour and 50% in next hour.
Average percentage growth in population

Geometric Mean

- **For Raw data**

The Geometric Mean (G.M) of a data set containing n observations is the nth root of the product of the values. If there are n observations representing different values $x_1, x_2, \dots, x_n$ of the variate X then G.M represented by G is calculated as

$$G = (x_1 \cdot x_2 \cdot \dots \cdot x_n)^{1/n}$$

- **For frequency distribution**

Let $x_1, x_2, \dots, x_n$ be the n values of the variate (in case of ungrouped frequency distribution) or values of class marks (in case of grouped frequency distribution) with corresponding frequencies $f_1, f_2, \dots, f_n$ where $\sum f = N = \text{Total Frequency}$. Then the G.M is calculated as

$$G = (x_1^{f_1} \cdot x_2^{f_2} \cdot \dots \cdot x_n^{f_n})^{1/N}$$

Example 1 :Find the G.M. of – 250, 12, 4.5, 119.5, 42, 35.4, 75, 30

Example 2: Find GM of following-

Marks	0-10	10-20	20-30	30-40	40-50
No of students	5	7	15	25	8

Weighted Arithmetic Mean

- Consider a data with N observations $X_1, X_2, X_3 \dots X_n$ with the weights assigned depending upon their significance or importance as $w_1, w_2, w_3 \dots w_n$, then AM is denoted by $\bar{X}$ and is defined as-

$$AM = \bar{X} = \frac{\sum wX}{\sum w}$$

$$= \frac{w_1X_1 + w_2X_2 + w_3X_3 + \dots + w_nX_n}{\sum w}$$

Example

Question: A student's grades in theory, practical and presentation in the subject graphics were 71, 78 and 89 respectively. If the credits of these grades are 2, 4 and 5 respectively, what is the appropriate average grade?

Solutions:

Combined Mean

It is used when we have to compute mean of two or more groups combined together.

If we have two groups of n_1 and n_2 items with arithmetic means $\bar{x}_1, \bar{x}_2$, then the Arithmetic Mean of the combined group of (n_1+n_2) items is given by

$$\bar{X} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$$

In general if there are n groups of $n_1, n_2, \dots, n_n$ items with respective means $\bar{x}_1, \bar{x}_2, \bar{x}_3, \dots, \bar{x}_n$, then the mean of the combined group is given by

$$\bar{X} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2 + n_3\bar{x}_3 + \dots + n_n\bar{x}_n}{n_1 + n_2 + \dots + n_n}$$

Example

Suppose a company works in 2 shifts The average weekly salary of 200 workers of morning shift is Rs1500 and the average weekly salary of 100 workers of night shift is 2100. Average salary of 300 workers

Total salary of 200 workers of morning shift $= 1500 * 200 = 300000$

Total salary of 100 workers of night shift $= 2100 * 100 = 210000$

Total salary of 300 workers $= 300000 + 210000 = 510000$

Average Salary $= 1700$

2. The mean monthly salary paid to 300 employees of a firm is Rs 14700. The mean monthly salary of 200 male employees is 15050. Find the mean monthly salary of remaining female employees

Harmonic Mean

- **Raw data**

If there are n observations $x_1, x_2, \dots, x_n$ of the variate X then their H.M. represented by H is calculated as the reciprocal of the A.M of the reciprocal of the values

$$H = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} + \dots + \frac{1}{x_n}} = \frac{n}{\sum(\frac{1}{x})}$$

- **Frequency data**

Let $x_1, x_2, \dots, x_n$ be the n values of the variate (in case of ungrouped frequency distribution) or values of class marks (in case of grouped frequency distribution) with corresponding frequencies $f_1, f_2, \dots, f_n$ where $\sum f = N = \text{Total Frequency}$. Then the G.M is calculated as

$$H = \frac{f_1 + f_2 + f_3 + \dots + f_n}{\frac{f_1}{x_1} + \frac{f_2}{x_2} + \frac{f_3}{x_3} + \dots + \frac{f_n}{x_n}} = \frac{\sum f}{\sum(\frac{f}{x})}$$

Four typists typed a letter in 4,5,8,and 10 minutes. Find out their average time taken to type the letter

X	1/X
4	0.25
5	0.20
8	0.125
10	0.10
Total	0.675

H = 5.93 minutes

Harmonic Mean

1. Cities A, B and C are equidistant from each other. A motorist travels from A to B at 30 mph, from B to C at 40 mph and from C to A at 50 mph. Determine his average speed.

$$\text{HM}=38.3$$

2. An aeroplane travels distances of 2500, 1200 and 500 km at speeds 500, 400 and 250 km/hr respectively. Find HM of the speed.

Distance = f

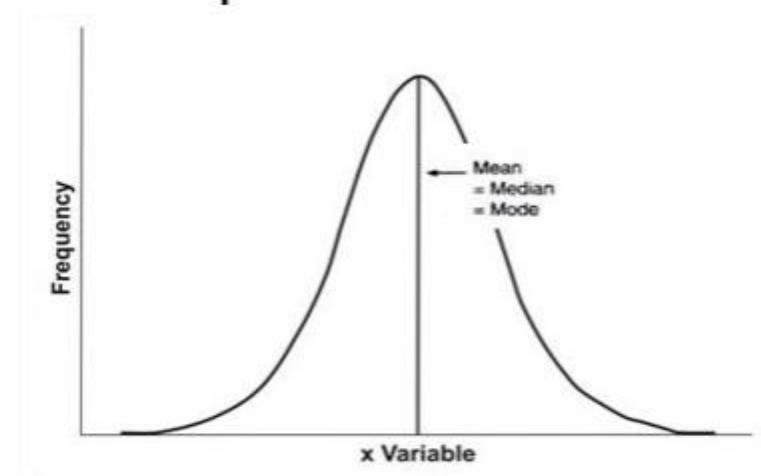
$$\text{HM}=420\text{km/hr}$$

Speed = x

Empirical Relation Between Mean Median and Mode

- If a frequency distribution graph is having a symmetrical frequency curve, the mean, median, and mode will be equal.

Mean= Median= Mode



- If a frequency distribution graph is having a asymmetrical frequency curve, then;

$$\text{Mode} = 3 \text{ Median} - 2 \text{ Mean}$$

OR

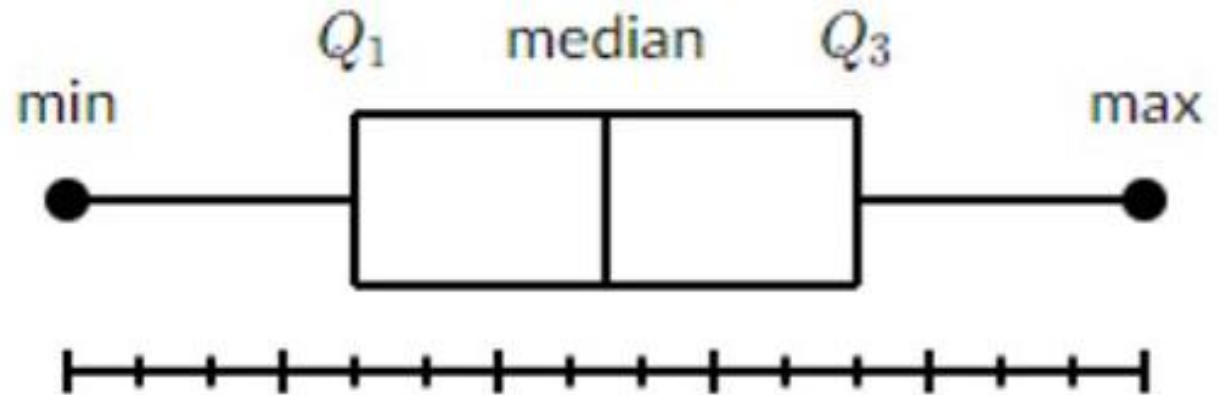
$$\text{Mean} - \text{Mode} = 3 (\text{Mean} - \text{Median})$$

BOX PLOT

1. It is also called as whisker plot
2. Displays the five-number summary of a set of data.

The five-number summary is:

- i. Minimum value of dataset
- ii. First quartile
- iii. Median
- iv. Third quartile
- v. Maximum value of dataset



EXAMPLE

Finding the five-number summary:

A sample of 10 boxes of raisins has these weights (in grams):

25, 28, 29, 29, 30, 34, 35, 35, 37, 38

Solution:

Step 1: Order the data from smallest to largest:

25, 28, 29, 29, 30, 34, 35, 35, 37, 38

Step 2: Find the minimum value of the dataset:

25

Step 3: Find the median:

$$\frac{30+34}{2} = 32$$

Step 4: Find the First quartiles:

The first quartile is the median of the data points to the left of the median.

$$Q_1 = 29$$

Step 5: Find Third quartiles:

The third quartile is the median of the data points to the right of the median.

$$Q_3 = 35$$

Step 6: Find the minimum value of the dataset:

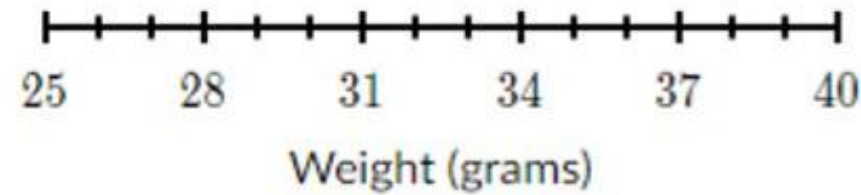
$$38$$

Step 7: Complete the five-number summary by finding the min and the max

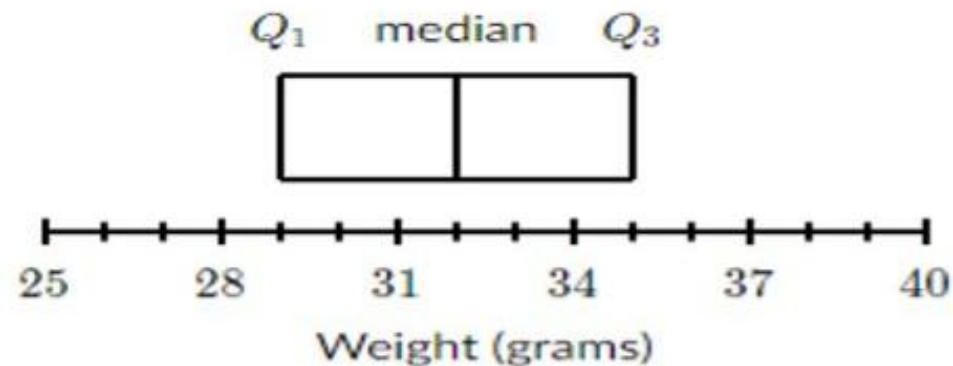
25, 29, 32, 35, 38

Making Box Plot:

Step 1: Scale and label an axis that fits the five-number summary.

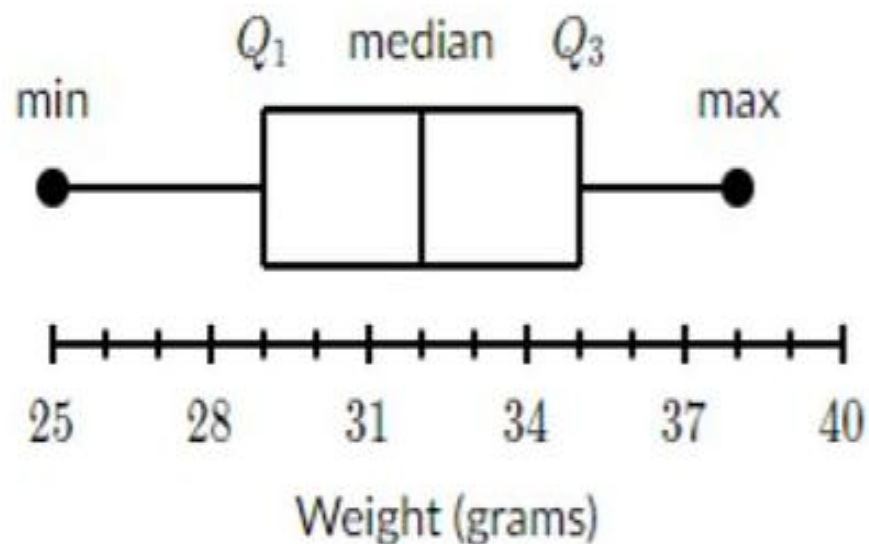


Step 2: Draw a box from Q_1 to Q_3 with a vertical line through the median.



Step 3: Draw a whisker from Q_1 to min *and from* Q_3 to the max.

Recall that the min is 25 and the max is 38



Mean

MERITS

- It is easy to compute and has a unique value.
- It is based on all the observations.
- It is well defined.
- It is least affected by sampling fluctuations.
- It can be used for further statistical analysis.

DEMERITS

- The mean is unduly affected by the extreme items (outliers).
- It cannot be determined for the qualitative data such as beauty, honesty etc.
- It cannot be located by observations on the graphic method.

MERITS

- It is easy to compute. It can be calculated by mere inspection and by the graphical method
- It is not affected by extreme values.
- It can be easily located even if the class intervals in the series are unequal

DEMERITS

- It is not amenable to further algebraic treatment
- It is a positional average and is based on the middle item
- It does not take into account the actual values of the items in the series

MODE

MERITS

- It is comparatively easy to understand.
- It can be found graphically.
- It is easy to locate in some cases by inspection.
- It is not affected by extreme values.
- It is the simplest descriptive measure of average

DEMERITS

- It is not suitable for further mathematical treatment.
- It is an unstable measure as it is affected more by sampling fluctuations.
- Mode for the series with unequal class intervals cannot be calculated.
- In a bimodal distribution, there are two modal classes and it is difficult to determine the values of the mode